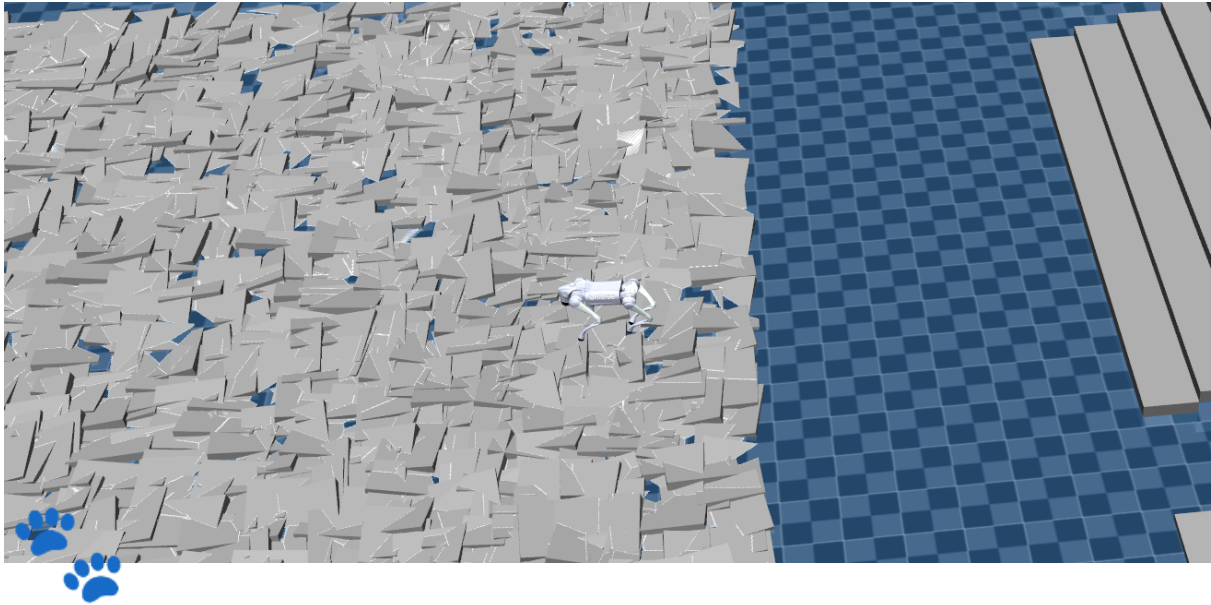




GO2 Sim2Sim & Sim2Real

🕒 Last updated time	April 19, 2026 3:41 PM
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1. ROS2 & Conda ENV Preperation

- ROS2 foxy or humble

```
# necessary packages, replace foxy with humble if using  
humble  
sudo apt install ros-foxy-rmw-cyclonedds-cpp  
sudo apt install ros-foxy-rosidl-generator-dds-idl  
sudo apt install libyaml-cpp-dev
```

- Conda environment with python 3.10

```
source ~/miniconda3/etc/profile.d/conda.sh  
conda create -n go2_deploy_ros2_env python=3.10  
conda activate go2_deploy_ros2_env
```

2. Package Build

Note: **DEACTIVATE CONDA BEFORE BUILDING !!!!!!!!!!!**

1. Clone repos:

- a. <https://github.com/Glowing-Torch/Deploy-an-RL-policy-on-the-Unitree-Go2-robot.git>
- b. https://github.com/unitreerobotics/unitree_ros2.git

```
# 1. git the main repo
git clone git@github.com:Glowing-Torch/Deploy-an-RL-policy-on-the-Unitree-Go2-robot.git

# 2. git unitree ros2 sdk in main repo
cd src
git clone git@github.com:unitreerobotics/unitree_ros2.git

# 3. git cyclonedds sdk in unitree_ros2
cd unitree_ros2/cyclonedds_ws
git clone https://github.com/ros2/rmw_cyclonedds -b foxy
# humble if using ROS2 humble
git clone https://github.com/eclipse-cyclonedds/cyclonedds -b releases/0.10.x
```

2. Build CycloneDDS

```
# pwd = ../go2_deploy/src/unitree_ros2/cyclonedds_ws
colcon build --packages-select cyclonedds

# output
Starting >>> cyclonedds
Finished <<< cyclonedds [8.49s]

Summary: 1 package finished [9.67s]
```

3. Build Unitree_ROS2

```

cd ..
# pwd = .../go2_deploy/src/unitree_ros2
colcon build

# output
Starting >>> cyclonedds
Starting >>> unitree_api
Starting >>> unitree_go
Starting >>> unitree_hg
Finished <<< unitree_api [8.71s]
Finished <<< cyclonedds [10.3s]
Starting >>> rmw_cyclonedds_cpp
Finished <<< unitree_hg [10.7s]
Finished <<< rmw_cyclonedds_cpp [4.45s]
Finished <<< unitree_go [16.2s]
Starting >>> unitree_ros2_example
--- stderr: unitree_ros2_example
...
---
Finished <<< unitree_ros2_example [14.1s]

Summary: 6 packages finished [30.4s]
1 package had stderr output: unitree_ros2_example

# error of unitree_ros2_example is not matter

```

4. Build Deploy_RL_Policy

```

cd ..
# pwd = .../go2_deploy
source .../go2_deploy/src/unitree_ros2/install/setup.zsh

colcon build --packages-select deploy_rl_policy

# output
Starting >>> deploy_rl_policy

```

```
Finished <<< deploy_rl_policy [8.62s]  
Summary: 1 package finished [8.75s]
```

3. Run MuJoCo Sim2Sim with XboxController

1. Dependencies install

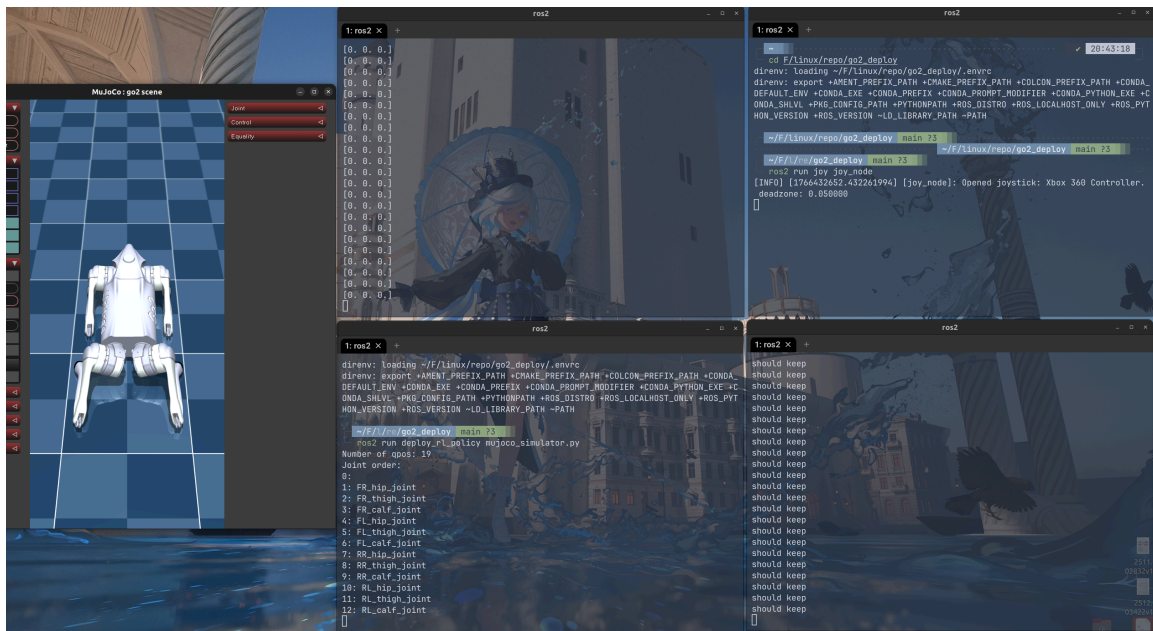
```
conda activate go2_deploy_ros2_env  
pip install torch mujoco
```

2. Recommended: direnv for env setup after packages building

```
source ~/miniconda3/etc/profile.d/conda.sh  
conda activate go2_deploy_ros2_env  
  
source /opt/ros/humble/setup.bash  
source ../go2_deploy/src/unitree_ros2/install/setup.bas  
h  
source ../go2_deploy/install/setup.bash
```

3. Run sim2sim node in split terminals:

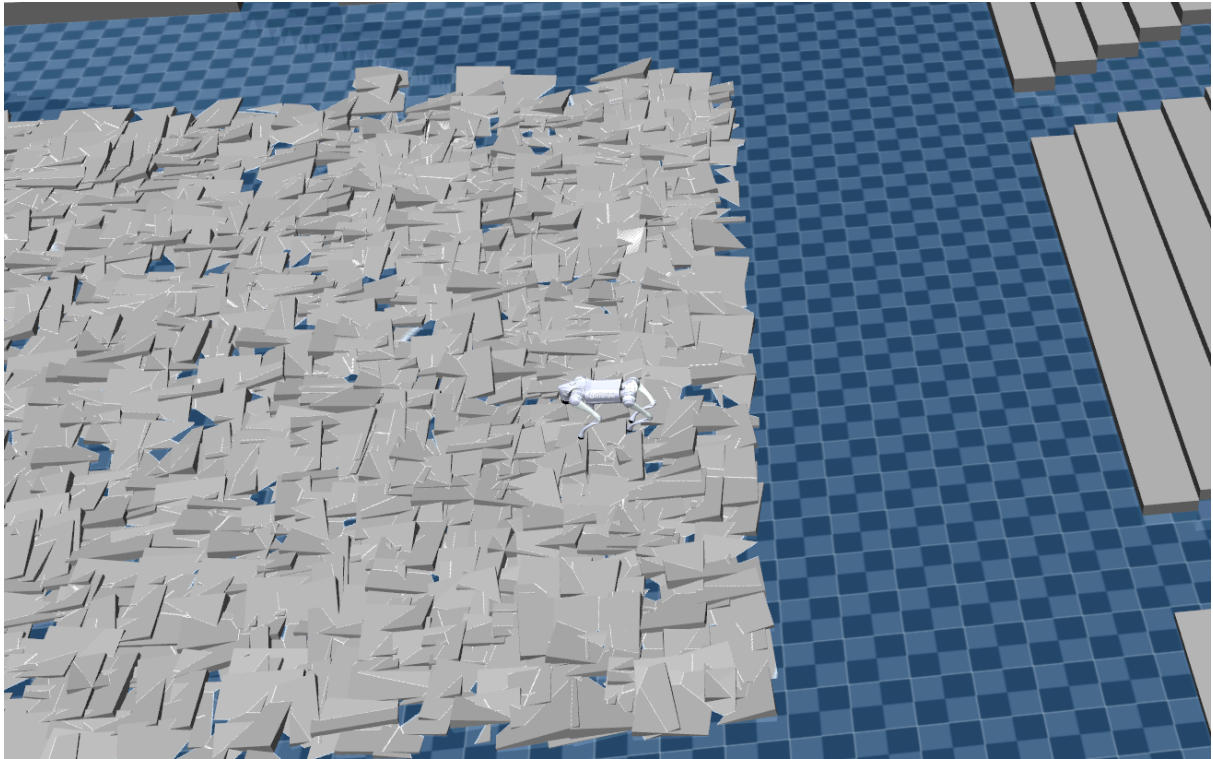
```
# 1. MuJoCo Simulator  
ros2 run deploy_rl_policy mujoco_simulator.py  
  
# 2. XBox Controller Interface  
ros2 run joy joy_node  
  
# 3. State Machine Controller, set is_simulation:=true f  
or sim2sim  
ros2 run deploy_rl_policy low_level_ctrl --ros-args -p i  
s_simulation:=true  
  
# 4. Reinforcement Learning Policy  
ros2 run deploy_rl_policy rl_policy.py --is_simulation T  
rue
```



4. Using Xbox controller to control the quadruped robot in MuJoCo:

- Initial State: Robot automatically enters "Laying Down" state
- B Button: Transitions from "Laying Down" → "Standing Up"
- A Button: Transitions from "Standing Up" → "Laying Down"
- LB + RB Simultaneously: While standing, executes RL Policy (remains in standing state)

Stay pressing LB + RB when you want to control robot by sticks. Then you can play it freely.



*Note: The controller will output the direction where you wanna robot move in the world frame by default policy (kp_kd)

4. Deploy Your Own Policy

5. Export Policy to Onboard Jetson and Run Sim2Real